

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 5 Microbial Growth and Its Control

5.1 Multiple Choice Questions

1) Which of the following is/are transferred to daughter cells during bacterial cell division?

- A) only genomic DNA, the blueprint for everything the cell needs
- B) individual genes and enzymes to later be brought together to form the genome
- C) only the chromosome and enzymes to immediately begin cellular metabolism
- D) chromosome, proteins, and all other cellular constituents

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.1

2) Bacterial growth refers to

- A) an increase in the number of cells.
- B) the occurrence of binary fission.
- C) the time required for one cell to divide into two.
- D) the occurrence of mitosis.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.1

3) The time interval required for the formation of two cells from one is called the

- A) generation time.
- B) growth time.
- C) growth rate.
- D) division rate.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.1

4) Turbidity measurements are commonly utilized for monitoring

- A) planktonic cultures.
- B) biofilms.
- C) colonies.
- D) sessile cultures.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.8

5) The time between inoculation and the beginning of growth is usually called the

- A) lag phase.
- B) log phase.
- C) dormant phase.
- D) death phase.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.3

6) Optical density and viable cell concentration are LEAST proportional to each other during the

- A) lag phase.
- B) exponential growth phase.
- C) stationary phase.
- D) death phase.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.8

7) When attempting to determine viable cell counts of a heat-sensitive bacterial isolate, which cell counting method should be avoided?

- A) spread-plate method
- B) pour-plate method
- C) live staining and direct counting with a microscope
- D) both spread- and pour-plate methods

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.7

8) Bacteria that are able to grow in humans and cause disease would be classified as

- A) psychrophiles.
- B) mesophiles.
- C) thermophiles.
- D) hyperthermophiles.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.9

9) An agar plate for counting colonies and maximizing statistical validity should ideally contain

- A) 1 to 100 colonies.
- B) 50 to 100 colonies.
- C) 30 to 300 colonies.
- D) 100 to 1000 colonies.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.7

10) Relative to enzymes in mesophilic microorganisms, which of the following is NOT characteristic of enzymes in psychrophiles?

- A) decreased alpha helix content
- B) decreased beta sheet content
- C) fewer hydrogen bonds
- D) fewer ionic bonds

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.10

11) The number of colonies obtained in a plate count does NOT depend on the

- A) inoculum volume.
- B) size of the colonies.
- C) type of culture medium.
- D) incubation time.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.7

12) Cell density in a chemostat is controlled by

- A) the concentration of the limiting nutrient.
- B) the dilution factor.
- C) the size of the initial inoculum.
- D) the concentration of the limiting nutrient, the dilution factor, and the size of the initial inoculum.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.4

13) What temperature is most commonly used in autoclaves to sterilize growth media and other devices prior to experimentation?

- A) 95°C
- B) 101°C
- C) 121°C
- D) 140°C

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.15

14) Most mesophilic organisms can grow in a temperature range of

- A) 0-15°C.
- B) 10-20°C.
- C) 20-45°C.
- D) 50-65°C.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.9

15) Obligate anaerobes which are sensitive to O₂ would be found growing

- A) throughout a tube of thioglycolate broth.
- B) only at the very top of a tube of thioglycolate broth.
- C) only at the bottom of a tube of thioglycolate broth.
- D) approximately one-third of the way down the thioglycolate broth.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.16

16) A microbe growing in a refrigerator is likely

- A) psychrophilic.
- B) mesophilic.
- C) psychrotolerant or psychrophilic.
- D) hyperthermophilic.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.10

17) Which of these statements is/are TRUE?

- A) In general, species that can grow at higher temperatures are prokaryotic.
- B) The most thermophilic prokaryotes are species of *Archaea*.
- C) Chemoorganotrophic organisms are able to grow at higher temperatures than phototrophic organisms.
- D) All of the statements are true.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

18) A chemical that denatures proteins is most likely to be classified as a(n) _____ agent.

- A) antiseptic
- B) bacteriostatic
- C) bacteriocidal
- D) detergent

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.17

19) The ratio of the vapor pressure of the air in equilibrium with a substance to the vapor pressure of pure water is known as

- A) water activity.
- B) vapor activity.
- C) positive water balance.
- D) osmosis.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.13

20) Organisms able to live in environments with high sugar concentrations are

- A) halotolerant.
- B) osmophiles.
- C) xerophiles.
- D) anaerobic fermenting bacteria.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.13

21) When water activity is low, an organism must

- A) increase its internal solute concentration.
- B) increase its external solute concentration.
- C) decrease its internal solute concentration.
- D) decrease its external solute concentration.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.13

22) A bacterium possessing _____ will produce O₂ bubbles when placed in the presence of hydrogen peroxide.

- A) catalase
- B) superoxide dismutase
- C) superoxide reductase
- D) peroxidase

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.14

23) A halotolerant facultative anaerobe would grow BEST in which of the following environments?

- A) oxygen depleted saline
- B) oxygenated saline
- C) oxygen depleted non-saline
- D) oxygenated non-saline

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.13

24) Which of the following forms of oxygen is/are generally toxic to living organisms?

- A) superoxide anion
- B) hydrogen peroxide
- C) hydroxyl radical
- D) Superoxide, hydrogen peroxide, and hydroxyl radicals are all toxic.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.14

25) Superoxide dismutase and catalase work together to convert superoxide into

- A) peroxide.
- B) oxygen.
- C) water
- D) water and oxygen.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.14

26) Which of the following methods used to enumerate cells often requires specialized staining to observe non-pigmented bacteria?

- A) spectrophotometry/turbidity
- B) spread-plating
- C) microscopy
- D) spread-plating, turbidity, and microscopy

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.6

27) Which term is most relevant in describing the efficacy of an antimicrobial for use in a clinical setting?

- A) effective dose
- B) sterilization coefficient
- C) lethal dosage
- D) minimum inhibitory concentration (MIC)

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.17

28) Which method would be LEAST effective at sterilizing a glass hockey stick to use in the spread-plate method?

- A) autoclaving
- B) gamma radiation
- C) ethanol soaking
- D) ultrahigh-temperature pasteurization

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.15

29) When counting colonies on an agar plate

- A) it is assumed that each colony arose from division of one or a few cells.
- B) only viable colonies are counted.
- C) the medium must be suitable for colony growth.
- D) All of these answer choices are true.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.5

30) By controlling the concentration of nutrients continuously added to a chemostat, cells can constantly be maintained at the

- A) exponential growth phase.
- B) stationary growth phase.
- C) lag growth phase.
- D) death growth phase.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.4

31) Serial dilutions are generally made

- A) when the appropriate viable count is unknown.
- B) based on powers of 10, although other serial dilution factors are possible.
- C) to reach a suitable dilution while minimizing error.
- D) All of these answer choices apply.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.7

32) Which of the following statements is FALSE?

- A) In the death phase, bacterial growth may cease as a result of oxygen and nutrient depletion.
- B) The generation time of bacteria may vary from species to species.
- C) Bacteria typically divide by binary fission, producing two daughter cells.
- D) In the lag phase, cell death exceeds cell division.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.3

33) The optimal growth temperature of a bacterium is most closely related to the optimal temperature for

- A) transcription of DNA.
- B) DNA replication.
- C) enzyme function.
- D) mRNA attachment to ribosomes.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.9

34) Why is the enzyme catalase important to the survival of bacteria?

- A) Catalase protects the cell from desiccation.
- B) Catalase breaks down toxic hydrogen peroxide.
- C) Catalase allows for growth in high salt concentrations.
- D) Catalase aids in the transport of oxygen into the cell.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.14

35) A medium containing known substances in precise amounts is known as a _____ media.

- A) selective
- B) complex
- C) natural
- D) defined

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.5

36) Selective medium differs from differential medium because

- A) selective medium permits growth of more organisms than differential medium.
- B) selective medium permits growth of a particular microbial type while differential medium is used to distinguish between types of organisms.
- C) differential medium differentiates pathogens from nonpathogens while selective medium grows only pathogens.
- D) differential medium contains growth inhibitors while selective medium does not.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.5

37) The cell membranes of thermophiles and hyperthermophilic bacteria typically have

- A) more saturated fatty acids.
- B) more long chain fatty acids.
- C) fewer unsaturated fatty acids.
- D) All of these answer choices are true of thermophiles and hyperthermophilic bacteria.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

38) The use of chemical agents directly on exposed body surfaces to destroy or inhibit vegetative pathogens is

- A) disinfection.
- B) sterilization.
- C) antisepsis.
- D) sanitization.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.17

39) Which of the following statements is TRUE of disinfectants?

- A) They are always effective in destroying endospores.
- B) They are used on living tissue.
- C) They are used for sterilization.
- D) They are used on inanimate surfaces.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.17

40) A _____ agent is a chemical that inhibits bacteria from reproducing, but does NOT necessarily kill them.

- A) bacteriostatic
- B) bacteriolytic
- C) bacteriocidal
- D) xerophilic

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.17

41) Microbial contamination is prevented by the

- A) use of the quadrant streak method.
- B) use of aseptic technique.
- C) use of cultures containing special nutrients.
- D) spread plate method.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.5

42) A common antimicrobial chemical (or chemicals) used as both an antiseptic and a general disinfectant is

- A) iodophors.
- B) 70% alcohol.
- C) hydrogen peroxide.
- D) Each of these chemicals at appropriate concentrations can be used as antiseptics or disinfectants.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.17

43) Cultures of a bacterial species were incubated on the shelf of a refrigerator (5°C), on a lab benchtop (25°C), on the shelf of a 37°C incubator, and on the shelf of a 50°C incubator. After incubation, there was no growth at 37°C or 50°C, slight growth on the benchtop, and abundant growth at refrigeration. What term could be used for this species?

- A) halophile
- B) mesophile
- C) anaerobe
- D) psychrophile

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.9

44) A microbiology student in a research lab repeated an experiment described by others using the identical bacterial isolate, the same growth medium, and the same growth conditions. However, the student was unable to achieve the same O.D. at 600 nm. What is the LEAST likely cause for this discrepancy in measured turbidity?

- A) One experiment used 16 mm wide test tubes while the other used 18 mm wide test tubes.
- B) One experiment subtracted the yellow color of the growth medium away from the final turbidity reported whereas the other used colorless water.
- C) One experiment vigorously dispersed the bacterial pellicle with vortexing while the other did not.
- D) The two experiments were carried out at vastly different elevations and did not consider the influence of pressure.

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.8

5.2 True/False Questions

1) Microbial growth is generally described as an increase in cell number rather than the expansion in size of an individual microbial cell.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.1

2) The duration of exponential growth would increase if bacterial cells divided into three equal daughter cells rather than two.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.3

3) A selective medium is used to suppress the growth of certain bacteria in order to facilitate the growth of other bacteria.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.5

4) The lag phase does NOT occur if an exponentially growing culture is transferred into a nutrient rich medium with the same components and growth conditions as the previous medium.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.3

5) The rate of exponential growth varies greatly according to bacterial species and bioavailable nutrients.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.3

6) A bacterium whose optimal growth temperature is 35°C would be classified as a thermophile.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.3

7) A complex medium is one that has been completely made in the laboratory and the concentration of all the nutrients is known.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.5

8) Many fungi and bacteria grow best at pH 5 or below.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.12

9) In both the lag and stationary phase, there is no net increase or decrease in viable cells.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.3

10) The death phase applies to individual cells rather than populations.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.3

11) Direct microscopic counting of cells is an accurate method for estimating the number of viable cells in a sample.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.7

12) The optimal pH for growth of an organism refers to its intracellular environment.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.12

13) When viable cell concentrations are too high to count on an agar plate, it is common to use larger sized plates to increase the surface area for counting the colonies.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.7

14) Refrigeration controls microbial growth in food as a result of irreversible cell damage caused by low temperature.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.9

15) The optimum growth temperature for an organism is typically closer to its minimum temperature rather than the maximum temperature.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.9

16) A bacterium such as a snow alga that is able to grow in cold temperatures is called a psychrophile.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.9

17) Some microbes can grow in boiling water.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

18) Organisms living in boiling hot springs often grow rapidly and have short doubling times.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

19) *Taq* polymerase was isolated from a thermophile and is used in the polymerase chain reaction (PCR) technique because it does not become inactivated at high temperatures.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

20) Mesophiles typically have longer fatty acid tails and more saturated carbon-carbon bonds in their cytoplasmic tails compared with thermophiles.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.11

21) Knowledge of microbial growth patterns is useful in working with the control of microbial growth.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.15

22) Irreversible cell damage is more likely to occur at low rather than high temperatures.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.9

23) Based on studies with obligate acidophiles, high concentrations of hydrogen ions are required for their membrane stability.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.12

24) A bacterial cell in nutrient broth with a generation time of 15 minutes will produce 16 cells in 2 hours.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.2

25) Planktonic cells remain in suspension while sessile cells adhere to a surface.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.1

26) The thermal death time is the time needed to kill all the bacteria in a particular culture at a certain temperature.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.15

27) Autoclaving and pasteurization are two processes that are both very effective for sterilization.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.15

28) Most microorganisms are particularly susceptible to antimicrobial agents during the logarithmic growth phase.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.17

29) Sterility of a laminar flow hood is accomplished by filter-sterilized air passed through the hood quickly enough that non-sterile air does not flow into the work area.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.16

5.3 Essay Questions

1) Use the formulas $N = N_0 2^n$ and $g = t/n$.

N = final cell number, N_0 = initial cell number,

n = number of generations, t = time, g = generation time.

Find the generation time if $N = 2 \times 10^8$, $N_0 = 3 \times 10^6$ and $t = 3$ hours.

Answer: The number of generations (n) must first be determined through the rearranged formula: $n = [(\log N) - (\log N_0)] / (\log 2)$ to determine $n = 6.059$ generations. Finally, the generation time (g) = t/n , which equates to nearly 0.5 hours per generation or almost 30 minutes per generation.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.2

2) Explain the difference between the division rate and the generation time.

Answer: Generation time (g) and division rate (v) are reciprocal values of each other; both are determined by monitoring exponential growth over time. Division rate is reported as the number of generations per time and is calculated by taking $1/g$. The units for g are in time per generation, which is the doubling time during exponential growth.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.2

3) Knowing the concentration of microorganisms in a sample is often an important consideration for environmental, industrial, and medical microbiologists, yet a microscope is sometimes not used to accomplish this task. Explain five major limitations to using a light microscope to directly count microorganisms.

Answer: All of the following eight are potential issues associated with the direct counting method: 1) live and dead cells are indistinguishable without special staining, 2) low cell concentrations are challenging to enumerate, 3) motile cells must be immobilized, 4) precision can be quite poor with manually counting hundreds or thousands of cells, 5) small cells are difficult to observe and count even at 1,000X total magnification, 6) small particulates can be mistaken for microbial cells, 7) unstained cells require more sophisticated microscopy, and 8) it is very time-consuming.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.6

4) Compare and contrast the spread plate method and the pour plate method of doing plate counts. Also describe which group of organisms would not be quantifiable in the pour plate method but would still be observed in the spread plate method.

Answer: Answers will vary, but one benefit to the pour plate method is that larger volumes of cells can be dispensed, whereas large liquid volumes dispensed on the surface of agar plates is often impractical. Many cells, especially psychrophiles and some mesophiles, cannot withstand the warm temperature of molten agar (~50°C) and therefore must be spread on top of an agar plate. Colonies are easier to enumerate on a two- rather than three-dimensional surface that favors the spread plate technique over the pour plate method.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.7

5) Enumerating the viable and total cell concentration of a population of a microbial isolate can be a laborious task for a microbiologist, especially when studying the same isolate for several years. It is often more practical to determine the relationship between optical density (OD) and cell concentration. Once this relationship (determined by a standard curve) is determined, the OD of an isolate in a broth can be used to determine the population's concentration. Why must a standard curve be prepared for each isolate when using OD measurements to determine cell concentration? Also describe an experiment that would generate this type of standard curve.

Answer: Answers will vary, but it should be noted that a particular OD value will correspond to a specific cell concentration due to different shapes and sizes of microorganisms. One such example could be a culture of *Citricella* sp. SE45 which was determined to have an OD₅₄₀ of 0.485. This OD fit into the already-constructed standard curve to indicate that 5×10^5 viable cells are in each milliliter of broth, which can be helpful in experiments where the final cell concentration or number is critical. To initially create the standard curve, a traditional growth curve experiment could be performed over time where both OD and cell concentration (viable with viability plating or total with direct counts) would be measured. The values of OD and cell concentration for each time point would then be used to create a standard curve, and a linear regression curve would show the relationship of OD to cell concentration such that subsequent experiments OD could be measured as a proxy for cell concentration.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.8

6) Explain why a hyperthermophile is unlikely to be a human pathogen.

Answer: The human body's temperature is approximately 37°C, which is not a favorable environment for a hyperthermophile, which has an optimal growth temperature of 80°C or higher.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

7) Explain the present understanding of molecular adaptations to the cytoplasmic membrane as found in psychrophiles.

Answer: Non-psychrophiles do not thrive in the cold environment, in part due to predominantly having large or saturated fatty acid chains in their cytoplasmic membrane, which become increasingly rigid and wax-like with colder temperatures. The psychrophiles have higher relative concentrations of short and unsaturated fatty acids in their membrane and therefore maintain a semi-permeable membrane when exposed to less heat.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.10

8) Explain the present understanding of molecular adaptations to the cytoplasmic membrane that are present in thermophiles.

Answer: In thermophiles and most hyperthermophilic *Bacteria*, the cytoplasmic membrane has a higher content of long-chain and saturated fatty acids and a lower content of unsaturated fatty acids than are found in the cytoplasmic membranes of mesophiles. Saturated fatty acids form a stronger hydrophobic environment than do unsaturated fatty acids, and longer-chain fatty acids have a higher melting point than shorter-chain fatty acids; collectively these properties increase membrane stability.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.11

9) Would you expect a xerophilic organism to be halotolerant? Why or why not?

Answer: Yes, xerophiles are typically halotolerant. For example, the xerophilic halotolerant fungus *Xeromyces bisporus* inhabits cereal, candy, and dried fruit, all of which are very dry environments that also contain salt. With all else being equal, a decrease in solvent (such as water) translates to an increase in solute (salt) concentration, and therefore a microbe capable of tolerating a low water activity likely can also tolerate higher salt concentrations.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.13

10) Describe the autoclave, and explain the roles of time, temperature, and pressure in the functioning of the autoclave.

Answer: High pressure is used in an autoclave so that liquids do not boil at temperatures beyond 100°C. The heat (not pressure) is used to kill living cells and is usually used for 15 minutes at 121°C. The time must be increased for bulky samples to ensure the objects or liquids are heated to 121°C for at least 15 minutes.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 5.15

11) Discuss the pros and cons of using radiation to sterilize items such as foods, drugs, and surgical supplies.

Answer: UV radiation sterilizes only the surfaces of objects which can be problematic for foods that are not fully cooked on the inside. Ionizing radiation can penetrate so they are ideal for foods such as grain cereals and ground beef. The high costs associated with the specialized equipment make it feasible only for large industrial applications. Due to the high energy of some of these radiation procedures, there are many concerns of potential radioactive contamination, production of carcinogenic or toxic products, and nutritional value alteration.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.16

12) Why is 100% ethanol less effective as an antimicrobial compared to a 70-95% concentration of ethanol?

Answer: 70% percent alcohol is ideal as opposed to a stronger solution. Pure alcohol coagulates protein on contact. If 100% alcohol is poured over a single-celled organism, the alcohol goes through the cell wall of the organism in all direction and coagulates the proteins found in the cell membrane. The coagulated membrane proteins would then stop the alcohol from penetrating further into the cell, preventing denaturation of cytoplasmic proteins. If this happened, the cell would become inactive but not dead. Under favorable conditions, the cell would then begin to function. If 70 percent alcohol is poured over a single celled organism, the diluted alcohol also coagulates the protein, but at a slower rate, so that it is able to penetrate all the way through the cell before protein coagulation can block it. Thus, the protein found in the entire cell is coagulated and the organism dies.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 5.17

13) Why are microbial doubling times in nature typically longer than those obtained in the laboratory?

Answer: The general theme should be on how lab conditions are designed to optimally grow a microorganism (e.g., pH, substrate(s), temperature, vitamins) and minimize pressure. In the environment this microbe often is in conditions that are less than ideal, where they likely will encounter competition for growth substrates and other essential chemicals.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 5.2